

Standard Discrete Distributions: The Discrete Uniform

We begin our exploration of standard distributions with the most fundamental concept of fairness: a system where every outcome in a finite set is equally likely.

The Discrete Uniform Distribution

Let a random variable X take on any integer value in a contiguous range from a to b inclusive.

- The total number of possible outcomes is $N = b - a + 1$.
- Because no outcome is favored, the probability is distributed evenly across the entire support.

Probability Mass Function (PMF)

We write $X \sim \text{DiscreteUniform}(a, b)$. The PMF is:

$$P(X = k) = \frac{1}{N} = \frac{1}{b - a + 1}, \quad \text{for } k = a, a + 1, \dots, b$$

Discrete Uniform: Expectation and Variance

1. Expectation (The Midpoint)

Using the definition of expectation and factoring out the constant probability:

$$E[X] = \sum_{k=a}^b k \frac{1}{N} = \frac{1}{N} \sum_{k=a}^b k$$

The sum of this arithmetic sequence is the average of the first and last terms multiplied by the total number of terms (N):

$$E[X] = \frac{1}{N} \left(N \frac{a+b}{2} \right) = \frac{a+b}{2}$$

Discrete Uniform Variance

Calculating the sum of squares directly from a to b is messy. Instead, we define a shifted variable $Y = X - a + 1$.

Y now takes values $1, 2, \dots, N$. Because shifting a distribution does not change its spread, $\text{Var}(X) = \text{Var}(Y)$.

1. Find the Second Moment $E[Y^2]$

Using the standard formula for the sum of consecutive squares:

$$E[Y^2] = \sum_{k=1}^N k^2 \frac{1}{N} = \frac{1}{N} \left[\frac{N(N+1)(2N+1)}{6} \right] = \frac{(N+1)(2N+1)}{6}$$

2. Calculate Final Variance

$$\begin{aligned} \text{Var}(Y) &= \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2} \right)^2 \\ \Rightarrow \text{Var}(Y) &= \frac{N+1}{2} \left[\frac{N-1}{6} \right] = \frac{N^2-1}{12} \quad \blacksquare \end{aligned}$$

The Continuous Uniform Distribution

We begin with the continuous equivalent of perfect fairness.

The Definition

A random variable $X \sim \text{Uniform}(a, b)$ is equally likely to take any real value in the continuous interval $[a, b]$. Because the density is evenly spread out, the PDF is a perfectly flat, horizontal line.

The PDF Formula

To ensure the total area under the curve equals exactly 1 (a rectangle of width $b - a$), the height must be $\frac{1}{b-a}$:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Continuous Uniform: Expected Value

Let $X \sim \text{Uniform}(a, b)$ with PDF $f(x) = \frac{1}{b-a}$. We find the expected value via integration:

$$\begin{aligned}
 E[X] &= \int_a^b xf(x)dx \\
 &= \int_a^b x \left(\frac{1}{b-a} \right) dx \\
 &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b \\
 &= \frac{1}{b-a} \left(\frac{b^2 - a^2}{2} \right)
 \end{aligned}$$

Factoring the difference of squares in the numerator, $b^2 - a^2 = (b-a)(b+a)$:

$$E[X] = \frac{1}{b-a} \frac{(b-a)(b+a)}{2} = \frac{a+b}{2} \quad \blacksquare$$

Continuous Uniform: Variance

First, we calculate the second moment, $E[X^2]$:

$$\begin{aligned} E[X^2] &= \int_a^b x^2 \left(\frac{1}{b-a} \right) dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b \\ &= \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(a^2 + ab + b^2)}{3(b-a)} = \frac{a^2 + ab + b^2}{3} \end{aligned}$$

Now, apply the variance formula $\text{Var}(X) = E[X^2] - (E[X])^2$:

$$\begin{aligned} \text{Var}(X) &= \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2} \right)^2 \\ &= \frac{a^2 + ab + b^2}{3} - \frac{a^2 + 2ab + b^2}{4} \end{aligned}$$

Finding a common denominator of 12:

$$\begin{aligned} \text{Var}(X) &= \frac{4a^2 + 4ab + 4b^2 - 3a^2 - 6ab - 3b^2}{12} \\ &= \frac{a^2 - 2ab + b^2}{12} = \frac{(b-a)^2}{12} \quad \blacksquare \end{aligned}$$

Generating Randomness: The Foundation

Computers are inherently deterministic, meaning they cannot generate truly random numbers. Instead, we rely on algorithms called **Pseudo-Random Number Generators (PRNGs)** to create sequences $U_1, U_2, \dots$ that behave like standard uniforms, $\text{Uniform}(0, 1)$.

What must a PRNG ensure?

To be mathematically valid for simulation, the algorithm must guarantee:

- **Uniformity:** The numbers must be spread evenly across $[0, 1]$. No sub-interval should be favored.
- **Independence:** Successive numbers must lack correlation. Knowing U_n should give you no predictable information about U_{n+1} .
- **Long Period:** Because the system has finite memory and it is deterministic, the sequence will eventually repeat. The length of this cycle (the "period") must be massive to avoid repeating patterns during a simulation.
- **Efficiency:** It must be computationally fast.

A Simple Algorithm: The Linear Congruential Generator

One of the oldest and most fundamental PRNG algorithms is the **Linear Congruential Generator (LCG)**. It generates a sequence of pseudo-random integers $X_0, X_1, X_2, \dots$ using a simple recursive formula:

$$X_{n+1} = (aX_n + c) \pmod{m}$$

Where the algorithm is defined by four integer parameters:

- $m > 0$: the **modulus** (which dictates the maximum possible period)
- $a \in \{1, \dots, m-1\}$: the **multiplier**
- $c \in \{0, \dots, m-1\}$: the **increment**
- $X_0 \in \{0, \dots, m-1\}$: the **seed** (the initial starting state)

To obtain the standard uniform output $U \sim \text{Uniform}(0, 1)$, we scale the result:

$$U_{n+1} = \frac{X_{n+1}}{m}$$

LCG: A "Toy" Example

Let's build a miniature LCG to see the algorithm in action.

The Setup

Let Modulus $m = 9$, Multiplier $a = 2$, Increment $c = 0$, and Seed $X_0 = 1$. The recurrence relation is: $X_{n+1} = (2X_n) \pmod{9}$

Generating the Sequence

- $X_1 = (2 \times 1) \pmod{9} = 2 \implies \mathbf{U_1 = 2/9 \approx 0.222}$
- $X_2 = (2 \times 2) \pmod{9} = 4 \implies \mathbf{U_2 = 4/9 \approx 0.444}$
- $X_3 = (2 \times 4) \pmod{9} = 8 \implies \mathbf{U_3 = 8/9 \approx 0.889}$
- $X_4 = (2 \times 8) \pmod{9} = 7 \implies \mathbf{U_4 = 7/9 \approx 0.778}$
- $X_5 = (2 \times 7) \pmod{9} = 5 \implies \mathbf{U_5 = 5/9 \approx 0.556}$

Note: Modern algorithms (like the Mersenne Twister) use highly complex bitwise operations and massive prime moduli to achieve periods like $2^{19937} - 1$, satisfying the strict criteria from our first slide.

Simulating General Uniform Distributions

How do we get from $U \sim \text{Uniform}(0, 1)$ to other uniform distributions? We use simple linear transformations.

1. Continuous Uniform on $[a, b]$

To generate $X \sim \text{Uniform}(a, b)$, we scale U by the width of our target interval $(b - a)$, and then shift it by the starting point a :

$$X = a + (b - a)U$$

2. Discrete Uniform on $\{a, a + 1, \dots, b\}$

To generate a random integer uniformly, we first create a continuous uniform variable over a slightly wider range $[a, b + 1)$, and then apply the floor function $\lfloor \cdot \rfloor$ to round down to the nearest integer:

$$X = \lfloor a + (b - a + 1)U \rfloor$$

Notice that the multiplier $(b - a + 1)$ is exactly N , the total number of distinct outcomes.

Generating from any CDF: The Inverse Transform Method

What if we want to simulate a random variable X from a distribution with a known Cumulative Distribution Function (CDF), $F(x)$?

The Generalized Inverse Transform Theorem

A valid CDF is non-decreasing but may have flat regions where a standard inverse is undefined. Instead, we use the **generalized (left) inverse**:

$$F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}$$

Let $U \sim \text{Uniform}(0, 1)$. If we define a new random variable as:

$$X = F^{-1}(U)$$

Then X is distributed exactly according to $F(x)$.

Generating from any CDF: The Inverse Transform Method

Proof: We want to show that the CDF of our new variable X is $F(x)$:

$$P(X \leq x) = P(F^{-1}(U) \leq x)$$

Because $F(x)$ is non-decreasing and right-continuous, the generalized inverse satisfies the fundamental equivalence $F^{-1}(u) \leq x \iff u \leq F(x)$. Therefore, we can rewrite the probability:

$$P(F^{-1}(U) \leq x) = P(U \leq F(x))$$

Since $U \sim \text{Uniform}(0, 1)$, we know $P(U \leq c) = c$ for any constant $c \in [0, 1]$. Because a valid CDF guarantees $0 \leq F(x) \leq 1$, it follows that:

$$P(U \leq F(x)) = F(x) \quad \blacksquare$$

Inverse Transform Method: Example

Example: The Exponential Distribution

Suppose X has CDF: $F(x) = 1 - e^{-\lambda x}$, for $x \geq 0$

- 1 Set a standard uniform variable U equal to the CDF:

$$U = 1 - e^{-\lambda X}$$

- 2 Solve algebraically for X to find the inverse $F^{-1}(U)$:

$$e^{-\lambda X} = 1 - U \implies -\lambda X = \ln(1 - U) \implies X = -\frac{1}{\lambda} \ln(1 - U)$$

- 3 Because $U \sim \text{Uniform}(0, 1)$, the quantity $(1 - U)$ is also exactly distributed as $\text{Uniform}(0, 1)$. For computational efficiency, we often simplify the algorithm to:

$$X = -\frac{1}{\lambda} \ln(U)$$

Inverse Transform Method: Discrete Example

Example: A Generic Discrete Distribution

Suppose X takes positive integer values $i = 1, 2, 3, \dots$ with probabilities $p_i = P(X = i)$. The CDF is a step function:

$$F(k) = P(X \leq k) = \sum_{i=1}^k p_i$$

Because the CDF is a step function, we rely entirely on the generalized inverse $F^{-1}(U) = \inf\{x : F(x) \geq U\}$. This essentially finds the first "step" of the CDF that reaches or exceeds U .

- 1 Generate a standard uniform variable $U \sim \text{Uniform}(0, 1)$.
- 2 Accumulate the probabilities $p_1, p_2, \dots$ until the sum is greater than or equal to U . In other words, find the smallest integer k such that:

$$U \leq \sum_{i=1}^k p_i$$

Inverse Transform Method: Discrete Example

A Concrete Discrete Distribution

Suppose X takes three values with the following probabilities (p_i) and CDF values $F(x) = P(X \leq x)$:

- $P(X = 1) = 0.2 \implies F(1) = 0.2$
- $P(X = 2) = 0.5 \implies F(2) = 0.2 + 0.5 = 0.7$
- $P(X = 3) = 0.3 \implies F(3) = 0.7 + 0.3 = 1.0$

Finding the Generalized Inverse $F^{-1}(u)$

Recall $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. To find the inverse, we see where a given probability $u \in (0, 1]$ lands in our CDF steps:

$$F^{-1}(u) = \begin{cases} 1 & \text{if } 0 < u \leq 0.2 \\ 2 & \text{if } 0.2 < u \leq 0.7 \\ 3 & \text{if } 0.7 < u \leq 1.0 \end{cases}$$

The Bernoulli Trial

The Bernoulli Trial

A Bernoulli trial is a random experiment with only two possible outcomes, typically labeled as "Success" (1) and "Failure" (0).

- Let p be the probability of success, where $0 \leq p \leq 1$.
- The probability of failure is therefore $1 - p$ (often denoted as q).

Examples:

- Flipping a biased coin (Heads = 1, Tails = 0).
- Testing a manufactured component (Pass = 1, Defective = 0).
- A bit transmitted over a noisy channel (Received correctly = 1, Error = 0).

The Bernoulli Distribution

If a random variable X represents the outcome of a Bernoulli trial, we write $X \sim \text{Bernoulli}(p)$.

The PMF can be compactly written for $x \in \{0, 1\}$ as:

$$P(X = x) = p^x(1 - p)^{1-x}$$

Using our fundamental definitions of expectation and variance:

- **Expectation:**

$$E[X] = \sum_{x \in \{0,1\}} x \cdot P(X = x) = (1 \cdot p) + (0 \cdot (1 - p)) = p$$

- **Second Moment:**

$$E[X^2] = (1^2 \cdot p) + (0^2 \cdot (1 - p)) = p$$

- **Variance:**

$$\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$$

Standard Discrete Models: The Binomial Distribution

What happens if we repeat a Bernoulli trial multiple times?

Definition via Summation

Let $X_1, X_2, \dots, X_n$ be n **independent** random variables, where each $X_i \sim \text{Bernoulli}(p)$.

We define a new random variable Y as their sum:

$$Y = \sum_{i=1}^n X_i$$

Y represents the total number of "Successes" in n independent trials. We say $Y \sim \text{Binomial}(n, p)$.

Examples:

- The number of Heads in 10 coin flips.
- The number of defective items in a batch of 50.

Binomial Probability Mass Function

To find the PMF of Y , we want the probability of getting exactly k successes (where $k \in \{0, 1, \dots, n\}$).

Deriving the Probability

- 1 **Specific Sequence:** The probability of getting k successes followed exactly by $n - k$ failures is $p^k(1 - p)^{n-k}$.
- 2 **Counting Combinations:** There are many ways to arrange k successes among n trials. The number of such unique sequences is given by the binomial coefficient $\binom{n}{k}$.

Multiplying these together yields the PMF:

$$P(Y = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Binomial Mean and Variance

We could calculate the expectation by solving $\sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}$, which requires complex algebra.

Instead, we use the linear properties we proved earlier!

1. Expectation (Linearity)

Because expectation is a linear operator:

$$E[Y] = E \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n p = np$$

2. Variance (Independence)

Because the trials X_i are **independent**, the covariance terms vanish, allowing us to distribute the variance operator:

$$\text{Var}(Y) = \text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) = \sum_{i=1}^n p(1-p) = np(1-p)$$

Binomial Expectation:

Let $Y \sim \text{Binomial}(n, p)$ and $q = 1 - p$. We calculate $E[Y]$ directly:

$$E[Y] = \sum_{k=0}^n k \binom{n}{k} p^k q^{n-k} = \sum_{k=1}^n k \frac{n!}{k!(n-k)!} p^k q^{n-k}$$

Cancel k with $k!$, leaving $(k-1)!$ in the denominator:

$$E[Y] = \sum_{k=1}^n \frac{n!}{(k-1)!(n-k)!} p^k q^{n-k}$$

Factor out np to reconstruct a new binomial coefficient:

$$E[Y] = np \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} q^{n-k} = np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} q^{(n-1)-(k-1)}$$

Let $j = k - 1$. By the Binomial Theorem, the remaining sum is $(p + q)^{n-1} = 1$:

$$E[Y] = np \cdot 1 = np \quad \blacksquare$$

Binomial Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[Y(Y-1)]$:

$$E[Y(Y-1)] = \sum_{k=0}^n k(k-1) \binom{n}{k} p^k q^{n-k}$$

The terms for $k=0$ and $k=1$ are zero. Starting at $k=2$, cancel $k(k-1)$ with $k!$:

$$E[Y(Y-1)] = \sum_{k=2}^n \frac{n!}{(k-2)!(n-k)!} p^k q^{n-k}$$

Factor out $n(n-1)p^2$ to rebuild the combinations:

$$E[Y(Y-1)] = n(n-1)p^2 \sum_{k=2}^n \frac{(n-2)!}{(k-2)!(n-k)!} p^{k-2} q^{n-k}$$

Just as before, the remaining sum is exactly $(p+q)^{n-2} = 1$. Thus:

$$E[Y(Y-1)] = n(n-1)p^2$$

Binomial Variance:

We just found $E[Y(Y - 1)] = n(n - 1)p^2$ and we know $E[Y] = np$.

1. Find the Second Moment $E[Y^2]$

Expand the factorial moment: $E[Y(Y - 1)] = E[Y^2 - Y] = E[Y^2] - E[Y]$.

Rearranging gives:

$$\begin{aligned} E[Y^2] &= E[Y(Y - 1)] + E[Y] \\ &= n(n - 1)p^2 + np \\ &= n^2p^2 - np^2 + np \end{aligned}$$

2. Calculate Final Variance

Apply the standard variance formula $\text{Var}(Y) = E[Y^2] - (E[Y])^2$:

$$\begin{aligned} \text{Var}(Y) &= (n^2p^2 - np^2 + np) - (np)^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= np - np^2 = np(1 - p) \quad \blacksquare \end{aligned}$$

The Sample Proportion ($\hat{p}$)

Instead of looking at the total successes Y , let's examine the **average number of successes** per trial, known as the sample proportion: $\hat{p} = \frac{Y}{n}$.

1. Expectation

$$E\left[\frac{Y}{n}\right] = \frac{1}{n}E[Y] = \frac{1}{n}(np) = p$$

The expected proportion of successes is exactly the probability of a single success.

2. Variance

Recall that pulling a constant out of a variance operator squares the constant:

$$\text{Var}\left(\frac{Y}{n}\right) = \frac{1}{n^2}\text{Var}(Y) = \frac{1}{n^2}np(1-p) = \frac{p(1-p)}{n}$$

The Sample Proportion: As $n \rightarrow \infty$

We just established that the variance of the sample proportion is inversely proportional to the number of trials:

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

The Infinite Limit

What happens mathematically if we repeat the trial an infinite number of times?

$$\lim_{n \rightarrow \infty} \text{Var}(\hat{p}) = \lim_{n \rightarrow \infty} \frac{p(1-p)}{n} = 0$$

Physical Implication: As n grows indefinitely, the sample proportion $\hat{p}$ stops fluctuating and collapses precisely onto the true, underlying probability p . This validates the frequentist definition of probability.

Sums of Independent Random Variables

Let X and Y be two **independent** random variables defined on the same probability space. We are interested in finding the probability distribution of their sum:

$$Z = X + Y$$

To determine the distribution of Z , we analyze all possible pairs (x, y) such that $x + y = z$. Because Y is constrained by $Y = z - X$, independence allows us to determine the distribution using the mathematical operation of **convolution**.

The Discrete Case: PMF Convolution

If X and Y are discrete independent random variables with PMFs $p_X(x)$ and $p_Y(y)$, the PMF of Z is obtained by summing over all valid disjoint supports:

$$p_Z(z) = P(X + Y = z) = \sum_x p_X(x)p_Y(z - x)$$

*This operation is frequently denoted as $p_Z = p_X * p_Y$.*

Convolution of Continuous Random Variables

The Continuous Case: PDF Convolution

If X and Y are continuous independent random variables characterized by the PDFs $f_X(x)$ and $f_Y(y)$, the density of their sum Z is evaluated via integration:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx$$

Property: Commutativity

Because algebraic addition is commutative ($X + Y = Y + X$), the convolution operator is symmetric. Thus, the order of the functions under the integral can be inverted:

$$\int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx = \int_{-\infty}^{\infty} f_Y(y)f_X(z-y) dy$$

Example: Discrete Support (Geometric Sum)

Problem Setup

Let X and Y be independent Geometric random variables representing the number of trials until the first success, both with success probability p . Their PMF for $k \in \{1, 2, 3, \dots\}$ is:

$$p_X(k) = p_Y(k) = (1 - p)^{k-1} p$$

Find the PMF of $Z = X + Y$.

Applying the Convolution Formula

Since the minimum value for both X and Y is 1, the minimum value for Z is $1 + 1 = 2$.

$$p_Z(z) = \sum_x p_X(x) p_Y(z - x)$$

The lower limit for x is 1. For $p_Y(z - x)$ to be non-zero, we also need $z - x \geq 1 \implies x \leq z - 1$.

Example: Geometric Sum (Solution)

Simplifying the Summation

We combine the terms with base $(1 - p)$ and group the p terms:

$$p_Z(z) = \sum_{x=1}^{z-1} p^2(1-p)^{x-1+z-x-1}$$

$$p_Z(z) = \sum_{x=1}^{z-1} p^2(1-p)^{z-2}$$

Final Result

Since the term inside the sum does not depend on x , we are simply adding a constant to itself $z - 1$ times:

$$p_Z(z) = (z-1)p^2(1-p)^{z-2}, \quad \text{for } z = 2, 3, 4, \dots$$

Note: This is exactly the Negative Binomial distribution $NB(2, p)$, which models

The "Rare Event" Limit

What happens to a Binomial distribution when the number of trials n becomes extremely large, but the probability of success p is extremely small?

The Setup

Consider modeling rare events in a massive population, such as:

- The number of typos in a 500-page book.
- The number of radioactive decays in a millisecond.

Let us take the limit as $n \rightarrow \infty$ and $p \rightarrow 0$. To keep the mathematics meaningful, we hold their product constant: let $np = \lambda$.

Here, λ represents the **average rate** of successes.

Deriving the Poisson PMF: Setup

Let $Y \sim \text{Binomial}(n, p)$. We substitute $p = \frac{\lambda}{n}$ into the Binomial Probability Mass Function and take the limit as $n \rightarrow \infty$:

$$\begin{aligned} \lim_{n \rightarrow \infty} P(Y = k) &= \lim_{n \rightarrow \infty} \binom{n}{k} p^k (1 - p)^{n-k} \\ &= \lim_{n \rightarrow \infty} \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \end{aligned}$$

We can group the constants (with respect to n) and pull them to the front:

$$= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} \left[\frac{n!}{(n-k)!} \cdot \frac{1}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k} \right]$$

Deriving the Poisson PMF: Evaluation

Let's break the expression inside the limit into three distinct components:

$$= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} \underbrace{\left[\frac{n(n-1) \dots (n-k+1)}{n^k} \right]}_{\text{Part A}} \underbrace{\left(1 - \frac{\lambda}{n} \right)^n}_{\text{Part B}} \underbrace{\left(1 - \frac{\lambda}{n} \right)^{-k}}_{\text{Part C}}$$

- 1 **Part A:** There are exactly k terms in the numerator polynomial and n^k in the denominator. As $n \rightarrow \infty$, the highest degree terms dominate, and this fraction approaches 1.
- 2 **Part B:** From standard calculus, the limit definition of the exponential function dictates that $\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n} \right)^n = e^{-\lambda}$.
- 3 **Part C:** For any finite k , as $n \rightarrow \infty$, the term inside the parentheses approaches 1. Thus, $1^{-k} = 1$.

The Poisson Distribution

Multiplying our evaluated limits together ($1 \cdot e^{-\lambda} \cdot 1$), we arrive at the PMF for a new, fundamental distribution.

The Poisson PMF

A random variable X follows a Poisson distribution with rate parameter $\lambda > 0$, denoted $X \sim \text{Poisson}(\lambda)$, if:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \text{for } k = 0, 1, 2, \dots$$

Mean and Variance of the Poisson: Intuitive

Because we derived this by holding $np = \lambda$ constant:

- **Mean:** $E[X] = \lim_{n \rightarrow \infty} (np) = \lambda$
- **Variance:** $\text{Var}(X) = \lim_{n \rightarrow \infty} np(1 - p) = \lambda(1 - 0) = \lambda$

Poisson Expectation: Direct Derivation

Let $X \sim \text{Poisson}(\lambda)$. By definition, the expected value is:

$$E[X] = \sum_{k=0}^{\infty} kP(X = k) = \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!}$$

The term for $k = 0$ is exactly zero, so we can start the sum at $k = 1$. We cancel k with $k!$, leaving $(k - 1)!$ in the denominator:

$$E[X] = \sum_{k=1}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k - 1)!}$$

We factor out $\lambda e^{-\lambda}$ to shift the index. Let $j = k - 1$:

$$E[X] = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k - 1)!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Recognizing the series expansion for $e^{\lambda} = \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$:

$$E[X] = \lambda e^{-\lambda} (e^{\lambda}) = \lambda \quad \blacksquare$$

Poisson Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[X(X-1)]$:

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1) \frac{\lambda^k e^{-\lambda}}{k!}$$

The terms for $k=0$ and $k=1$ are zero. Starting the sum at $k=2$, we cancel $k(k-1)$ with the $k!$ in the denominator:

$$E[X(X-1)] = \sum_{k=2}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-2)!}$$

Factor out $\lambda^2 e^{-\lambda}$ and let $j = k - 2$:

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Once again, the infinite sum resolves exactly to e^{λ} :

$$E[X(X-1)] = \lambda^2 e^{-\lambda} (e^{\lambda}) = \lambda^2$$

Poisson Variance: Direct Derivation

We now have the two necessary components: $E[X(X - 1)] = \lambda^2$ and $E[X] = \lambda$.

1. Find the Second Moment $E[X^2]$

Expand the factorial moment: $E[X(X - 1)] = E[X^2] - E[X]$.

Rearranging gives us:

$$\begin{aligned} E[X^2] &= E[X(X - 1)] + E[X] \\ &= \lambda^2 + \lambda \end{aligned}$$

2. Calculate Final Variance

Apply the standard variance formula $\text{Var}(X) = E[X^2] - (E[X])^2$:

$$\begin{aligned} \text{Var}(X) &= (\lambda^2 + \lambda) - (\lambda)^2 \\ &= \lambda^2 + \lambda - \lambda^2 \\ &= \lambda \quad \blacksquare \end{aligned}$$

Waiting Time: The Geometric Distribution

We return to our sequence of independent Bernoulli trials (where $P(\text{Success}) = p$ and $P(\text{Failure}) = q = 1 - p$).

Instead of counting the *number* of successes in a fixed time, we now ask:
How long must we wait until the very first success?

Defining the Random Variable

Let X be the number of trials needed to get the first success.

- The only way for $X = k$ to occur is to have exactly $k - 1$ failures followed immediately by 1 success.
- Because the trials are independent, we multiply their probabilities.

The Geometric PMF

We say $X \sim \text{Geometric}(p)$. Its Probability Mass Function is:

$$P(X = k) = q^{k-1}p, \quad \text{for } k = 1, 2, 3, \dots$$

Geometric Expectation:

By definition, $E[X] = \sum_{k=1}^{\infty} kP(X = k) = \sum_{k=1}^{\infty} kq^{k-1}p$.

We can pull the constant p outside the sum:

$$E[X] = p \sum_{k=1}^{\infty} kq^{k-1}$$

To evaluate this infinite series, we recall the standard geometric series for $|q| < 1$:

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$$

If we take the derivative of both sides with respect to q , the k drops down:

$$\frac{d}{dq} \sum_{k=0}^{\infty} q^k = \sum_{k=1}^{\infty} kq^{k-1} = \frac{d}{dq} (1-q)^{-1} = \frac{1}{(1-q)^2}$$

Substituting this result back into our expectation formula:

$$E[X] = p \left[\frac{1}{(1-q)^2} \right] = p \left[\frac{1}{p^2} \right] = \frac{1}{p} \quad \blacksquare$$

Geometric Variance and Intuition

To find the variance, we would use the second derivative of the geometric series to find the second factorial moment $E[X(X-1)]$.

Moments Summary

For $X \sim \text{Geometric}(p)$:

- **Mean:** $E[X] = \frac{1}{p}$
- **Variance:** $\text{Var}(X) = \frac{1-p}{p^2}$

Physical Intuition

The expected value $E[X] = 1/p$ aligns perfectly with human intuition:

- If you flip a fair coin ($p = 1/2$), you expect to wait $1/(1/2) = 2$ flips to see the first Heads.
- If you roll a fair die looking for a six ($p = 1/6$), you expect to wait $1/(1/6) = 6$ rolls to see the first 6.

The Memoryless Property

A unique and defining characteristic of the Geometric distribution is that it is **memoryless**.

Formal Definition

For any integers $s > 0$ and $t > 0$:

$$P(X > s + t \mid X > s) = P(X > t)$$

The probability of waiting t additional trials for a success, given that we have already waited s trials, is exactly the same as the probability of waiting t trials from the very beginning.

The samples do not "remember" past failures. Suppose you are rolling a fair die waiting for a 6.

- You have already rolled 10 times and failed.
- The die is not "due" for a 6. The probability that it takes 3 *more* rolls is exactly the same as taking 3 rolls on a brand new die.

Proof of the Memoryless Property

Prerequisite: Check that $P(X > k) = q^k$. (To fail to get a success in the first k trials means getting k failures in a row).

Using the definition of conditional probability:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t \text{ and } X > s)}{P(X > s)}$$

Since the event $\{X > s + t\}$ is a subset of the event $\{X > s\}$, their intersection is simply $\{X > s + t\}$. We substitute our tail probabilities:

$$\begin{aligned} P(X > s + t \mid X > s) &= \frac{P(X > s + t)}{P(X > s)} \\ &= \frac{q^{s+t}}{q^s} \\ &= q^t \\ &= P(X > t) \quad \blacksquare \end{aligned}$$

Conclusion: The past history of failures algebraically cancels out.

Exercise: Which Distribution?

Let's test our intuition. For each scenario, identify the most appropriate distribution model and its parameters.

1. The Single Seed

A plant seed has a 25% chance to sprout with green leaves. We plant exactly one seed. Let $X = 1$ if it is green, and $X = 0$ if it is not.

2. The Sales Calls

A worker makes 50 phone calls. Each call has an independent 5% chance of resulting in a sale. Let X be the total number of sales made today.

3. The Random Inspection

A computer randomly picks one shipping box out of 12 to inspect. Every box is equally likely to be picked. Let X be the chosen box number.

Exercise: Which Distribution

Identify the most appropriate distribution model and its parameters.

4. Summer Storms

Historical weather data shows that a specific town gets hit by lightning an average of 14 times per summer. Let X be the total number of lightning strikes this coming summer.

5. The Rare Blood Type

Only 2% of blood donors have a specific rare antibody. A doctor tests donated blood bags one by one. Let X be the total number of bags tested until they find the **very first** match.

Exercise

1. The Environmental Sensor

An oceanic sensor is designed to detect a specific microplastic particle. When a particle passes the sensor, there is a 10% chance the sensor successfully registers it.

- How many particles must pass by the sensor to ensure the probability of detecting **at least one** particle is strictly greater than 95%?

2. The Bakery Orders

A bakery receives an average of 2.5 large catering orders per week. Assume the weekly number of catering orders follows a Poisson distribution.

- **(a)** What is the probability the bakery receives at least 2 catering orders this week?
- **(b)** If we know the bakery received *at least one* order this week, what is the exact probability it received exactly 3 orders?

Exercise

3. The Tabletop Game

In a role-playing game, a player rolls a fair 20-sided die. They need to roll a "Natural 20" (a probability of $1/20$) to break a curse. They keep rolling until they succeed.

- What is the probability it takes **more than** 5 rolls to break the curse?

4. The Mystery Raffle

A charity raffle sells consecutively numbered tickets starting from Ticket #1 up to Ticket # N . The winning ticket is drawn with perfect fairness. If the **expected value** of the winning ticket number is 42:

- **(a)** How many total tickets (N) were sold?
- **(b)** What is the variance of the ticket numbers?

From Discrete to Continuous

Until now, our random variables jumped between distinct, separate values (counts, dice rolls, iterations). Now, we enter the continuous realm, where a variable can take **any real number** within a range (like exact weight, continuous time, or distance).

The Probability Density Function (PDF)

Instead of a Probability Mass Function (PMF), continuous variables are described by a PDF, denoted as $f(x)$.

- $f(x)$ is **not** a probability itself. It is a density.
- Probabilities are calculated by finding the **area under the curve** over a specific interval.

Because a continuous range has infinitely many points, the probability of hitting one exact, infinitely precise number is always zero:

$$P(X = c) = \int_c^c f(x)dx = 0$$

Deriving the Exponential: The Setup

Assume we have a continuous random variable $X \geq 0$ that possesses the memoryless property:

$$P(X > s + t \mid X > s) = P(X > t) \quad \text{for all } s, t \geq 0$$

Using the definition of conditional probability:

$$\frac{P(X > s + t \text{ and } X > s)}{P(X > s)} = P(X > t)$$

Since $(X > s + t)$ is a strict subset of $(X > s)$, their intersection is just $(X > s + t)$:

$$\frac{P(X > s + t)}{P(X > s)} = P(X > t) \implies P(X > s + t) = P(X > s)P(X > t)$$

To make this easier to read, let us define the **tail probability** (or survival function) as $G(x) = P(X > x)$. We now have a strict functional equation:

$$G(s + t) = G(s)G(t)$$

Deriving the Exponential: The Solution

We need to find a tail probability function $G(x)$ that satisfies our memoryless equation:

$$G(s + t) = G(s)G(t)$$

Finding the Function

The standard algebraic function that converts addition in its input to multiplication in its output is the exponential function:

$$e^{\lambda(s+t)} = e^{\lambda s} \cdot e^{\lambda t}$$

Because $G(x) = P(X > x)$ represents a probability, it has strict boundaries:

- It must start at $G(0) = 1$.
- It must strictly decrease as the waiting time x increases.

To make the function decay rather than grow infinitely, the exponent must be negative. Thus, for some rate parameter $\lambda > 0$:

$$G(x) = e^{-\lambda x}$$

The Exponential Distribution

We just proved that the tail probability must be $P(X > x) = e^{-\lambda x}$.

1. Find the CDF

The Cumulative Distribution Function, $F(x) = P(X \leq x)$, is simply the complement of the tail probability:

$$F(x) = 1 - P(X > x) = 1 - e^{-\lambda x}$$

2. Find the PDF

By the Fundamental Theorem of Calculus, the Probability Density Function $f(x)$ is the derivative of the CDF:

$$\begin{aligned} f(x) &= \frac{d}{dx} F(x) \\ &= \frac{d}{dx} (1 - e^{-\lambda x}) \\ &= -e^{-\lambda x} \cdot (-\lambda) \\ &= \lambda e^{-\lambda x} \quad \blacksquare \end{aligned}$$

Exponential Distribution: Mean Derivation

Let $X \sim \text{Exponential}(\lambda)$ with PDF $f(x) = \lambda e^{-\lambda x}$. By definition:

$$E[X] = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

We use Integration by Parts (IBP): $\int u dv = uv - \int v du$.

- Let $u = x \implies du = dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$\begin{aligned} E[X] &= [-xe^{-\lambda x}]_0^{\infty} - \int_0^{\infty} (-e^{-\lambda x}) dx \\ &= (0 - 0) + \int_0^{\infty} e^{-\lambda x} dx \quad (\text{Note: } \lim_{x \rightarrow \infty} \frac{x}{e^{\lambda x}} = 0 \text{ by L'Hôpital's}) \\ &= \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_0^{\infty} \\ &= 0 - \left(-\frac{1}{\lambda} \right) = \frac{1}{\lambda} \quad \blacksquare \end{aligned}$$

Exponential Distribution: Second Moment

To find the variance, we first need $E[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x} dx$. Using IBP again:

- Let $u = x^2 \implies du = 2x dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$\begin{aligned} E[X^2] &= [-x^2 e^{-\lambda x}]_0^\infty - \int_0^\infty (-e^{-\lambda x}) \cdot 2x dx \\ &= 0 + 2 \int_0^\infty x e^{-\lambda x} dx \end{aligned}$$

Notice that the remaining integral is almost $E[X]$. We multiply and divide by λ :

$$\begin{aligned} E[X^2] &= \frac{2}{\lambda} \int_0^\infty x \lambda e^{-\lambda x} dx \\ &= \frac{2}{\lambda} \cdot E[X] = \frac{2}{\lambda} \cdot \frac{1}{\lambda} = \frac{2}{\lambda^2} \end{aligned}$$

Exponential Distribution: Variance

Now we apply the variance formula:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

The Final Calculation

Substituting our derived moments:

$$\begin{aligned}\text{Var}(X) &= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 \\ &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} \\ &= \frac{1}{\lambda^2} \quad \blacksquare\end{aligned}$$

Summary for $X \sim \text{Exp}(\lambda)$

- **Mean:** $E[X] = 1/\lambda$
- **Standard Deviation:** $SD(X) = 1/\lambda$